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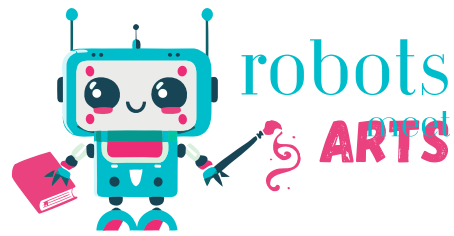
Project number: 2023-1-FR01-KA220-SCH-000151881

“ TREASURE HUNTERS: A ROBOTICS & STORYTELLING ADVENTURE ”



STIMMULI
for social change





Subject: Mother Language

Content: Robotics Fundamentals, Programming Concepts, Narrative Development, Script Writing, Theatrical Production, Inclusive Practices, Social-Emotional Learning Standards, Computer Science/Robotics, Arts, Visual & Performing

Computational practices

- **Decomposing**
- **Pattern recognition**
- **Abstraction**
- **Algorithms**
- **Problem solving**

Unplugged/plugged:
plugged



Preparation time:

30 min



Age:

9-11 years



Duration:

90 min



Location:

Classroom



Group distribution:

Two Groups of 4 students



Difficulty



After this lesson, the children will be able to...

Technical Skills: Design and construct a functional robot with specific movement capabilities. Write basic programs using SPIKE Prime App to control robotic movements. Document their technical process and design decisions.

Collaboration Skills: Work effectively in diverse teams with shared responsibilities. Negotiate roles and tasks fairly among group members. Integrate different ideas into a cohesive project. Give and receive constructive feedback. Express empathy and reflect critically on inequality.





Creative Expression: Generate original narrative content collaboratively. Translate abstract concepts into concrete storytelling elements. Perform expressively in front of an audience. Create visual art that supports storytelling.

Critical Thinking: Solve multi-step problems through systematic approaches. Connect technological tools with artistic expression. Adapt plans based on testing and feedback. Reflect on their learning process and outcomes.

Social-Emotional Development: Demonstrate empathy through character development and storytelling. Practice inclusive language and behaviors. Recognize diverse contributions to group projects. Build confidence in public speaking and presentation.

Materials needed:

- Lego Spike Essential kits (1 per group)
- Computers/tablets for programming (1 per group)
- Masking tape for creating floor paths/grids (1 per group)
- Chart paper and markers for brainstorming (1 set per group)
- Digital camera for documentation (1 per group)
- Markers, scissors, tape (just in case)
- Craft materials for props and scenery "Treasure" items (12-15 small objects, printed phrases, character images, playmobil items)
- Art supplies for prop/scenery creation
- Performance space
- Robot building (tables with ample workspace)
- Testing area (open floor space with room for robot movement)

Helpful links:

- <https://spike.legoeducation.com/>
- <https://www.youtube.com/watch?reload=9&v=ZkG1v1owSq4>
- <https://scratch.mit.edu/educators/>
- <https://www.bbc.co.uk/bitesize/guides/zp92mp3/revision/1>
- <https://www.readwritethink.org/classroom-resources/student-interactives/story>
- <https://www.teachstarter.com/us/blog/drama-games-for-kids-years-4-7-us/>

Preparation and teacher notes

Technology Preparation

Install LEGO Spike App on all devices. Create student accounts if using online platforms. Test all robot kits to ensure functionality. Charge all LEGO hubs before first session. Bookmark relevant tutorial websites and resources. Create a troubleshooting guide for common issues. Prepare a simple demonstration robot. Create student groups in advance (diversity of skills / abilities). Print building instructions or have digital access ready. Display vocabulary terms: algorithm, sequence, input, output.

Phase 1: Robot Engineers (50')

Ice-Breaker Activity:

Students work in pairs. They have to complete the sentence: "If I were a robot, I would be programmed to _.". Each student say their robot "purpose" aloud.

Session Outline:

Teachers begin by showing example videos of similar robotics projects to spark inspiration and set context. They explain the connection between technology and storytelling, highlighting how robotics can bring narratives to life.

Next, teachers help students establish team roles, which may rotate throughout the project: Builder, Programmer, Tester, and Documenter. LEGO kits and building guides are distributed. Teams construct a basic robot chassis (the taxi model) equipped with motors, while teachers circulate to assist with any technical difficulties. Once construction is complete, teams store their robots safely.



A brief reflection wraps up the session with guiding questions such as: "What was challenging?" and "What worked well?" Finally, teachers provide a preview of the next session, which will focus on programming.

Differentiation notes:

- For students with fine motor skill challenges, teachers pre-assemble particularly complex components.
- Advanced builders are offered optional design enhancements for added challenge.



Preparation and teacher notes

Technology Preparation

Prepare programming challenge cards of varying difficulty (seek for some examples in annex).

Set up simple test courses with tape on the floor.

Print / display programming vocabulary reference sheet (see example in Annex).

Phase 1: Robot Engineers (50')

Programming Session :

Teachers demonstrate the SPIKE Prime App and show how to connect it to the LEGO SPIKE system. They explain the different block categories and introduce basic commands. A simple program is modeled and tested live to illustrate the programming process.

Teams then connect their robots to devices and follow along as teachers guide them through creating a basic movement sequence. Robots are tested on the floor to verify functionality.

Students are then challenged with a series of progressively complex programming tasks, including:

- Move forward for 2 seconds
- Create a square path
- Navigate to a specific point and return

Differentiation notes:

- Command reference cards are provided to support visual learners.
 - Paired programming is encouraged, especially for students who may be struggling, with attention to balanced gender participation.
 - Extension challenges are available for students who finish early and seek additional engagement.
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Preparation and teacher notes

Gather "treasure" items representing: objects (small toys, craft items, colored blocks).

Phrases (printed on cards - proverbs, poetry lines, see example in Annex).

Prepare floor space for treasure map layout.

Create example station designs.

Phase 2: Treasure Hunt (15 minutes)

Teachers introduce the concept of the treasure hunt and explain the rules clearly. They show example treasures and discuss how these objects can play a role in storytelling. Some objects examples that can be used are: a hat, a teddy bear, a pair of sunglasses, a dustpan and a brush, a clay lamp, a rose and everything else that can be relates in order to make a story.



Teams then design the layout of their treasure map directly on the floor and place treasures at designated stations. Each team plans a robot navigation strategy to reach their assigned locations. Students program initial routes to specific stations, testing and refining their robot's movement for precision. Programming adjustments are made as needed, with successful approaches documented by the teams.



Preparation and teacher notes

Phase 2: Treasure Hunt (15 minutes)

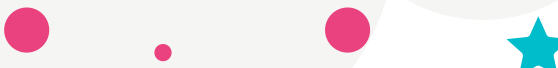
Robots navigate to the stations to "collect" treasures, which are recorded on collection cards. Between attempts, teams analyze results and make further refinements to improve accuracy.



At the end of the phase, teams display their collected treasures and briefly describe the strategies they used for navigation. A group discussion begins on how the different treasures could be connected into a shared or individual story.

Differentiation notes:

- Flexible time limits are provided for teams who need additional attempts.
- Teachers offer physical assistance with robot retrieval when necessary.



★ Preparation and teacher notes

Create story element templates.

Prepare example stories showing integration of diverse elements.

Set up "story inspiration" stations with books, images, music.

Have digital or paper story mapping tools available.

Prepare narrative feedback questions. Have examples of theater scripts.

Phase 3: Story Creation (15')

Teachers lead a discussion on the elements of compelling stories—such as object, setting, conflict, and resolution. They explain how to integrate random elements (like the collected treasures) into a cohesive narrative. Examples of creative integration techniques are shared to inspire students. Students may decide whether they want to integrate the robot in their scripts or if they want to set it aside.

Each team examines its collected treasures, brainstorming possible meanings and connections. Students sort their items by potential story role (e.g., object, setting, plot device). Using these ideas, teams begin drafting story concepts that incorporate the treasures.

Character sketches and setting descriptions are created to build the foundation of the narrative. Teams share their initial ideas with one another, giving and receiving feedback to refine their concepts.

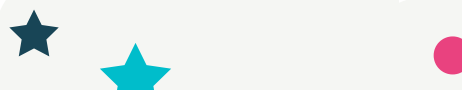
In this phase, it's very important that the teacher encourages the students to use their mother tongue correctly, urging them to employ rich vocabulary, subordinate clauses, many adjectives, and poetic language.

Differentiation notes:

- Provide story structure templates to guide students who need additional support.
- Offer voice recording options for students who struggle with writing.
- Create visual storyboards to support spatial/visual learners.

Teams then translate their story outlines into script format, determining narration needs and assigning speaking roles. A guided discussion is held around representing diverse perspectives and reviewing scripts for inclusive language and respectful representation.

Teams read portions of their scripts aloud and receive peer feedback focused on clarity, emotional impact, engagement and correct use of mother language.





Preparation and teacher notes

Organize art supplies for prop /set creation.

Set up creative work stations.

Prepare rehearsal schedule and spaces.

Phase 4: Theatrical Production (15')

Teachers introduce the key roles involved in theatrical production and explain their responsibilities. They demonstrate basic performance techniques such as voice projection, gesture, and staging.

Students then select their production roles, choosing from:

- Actors/Performers
- Designers (set, costume, props)
- Sound/Light Coordinators
- Directors

This phase marks the transition from script to performance, where students begin preparing for the dramatized presentation of their robotics-enhanced story.

Teams begin organizing their tasks collaboratively:

- Create lists of needed costumes and materials.
- Sketch simple set designs.
- Plan how robots will be integrated into the performance (if they choose to).
- Begin storyboarding key scenes visually to guide staging and timing.

Differentiation notes:

- Ensure roles align with students' strengths and interests.
 - Provide visual task cards for each production role to clarify responsibilities.
 - Offer examples and templates for designing sets, costumes, and props.
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Preparation and teacher notes

Phase 4: Theatrical Production (15')

Students then begin producing the necessary materials:

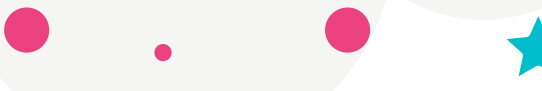
- Build simple props, costumes, and background elements.
- Design and test robot integration during specific moments in the performance.
- Develop appropriate sound effects or musical cues.
- Coordinate robot movement to synchronize with human actors, ensuring smooth transitions and storytelling flow.

Differentiation notes:

- Provide visual or written performance cue cards for students who benefit from step-by-step prompts.
- Establish and communicate calm-down procedures for students experiencing performance anxiety.

This phase emphasizes collaboration, creativity, and technical coordination, setting the stage for the final presentation of their integrated robotics-and-storytelling performance.





Preparation and teacher notes

Phase 5: Presentation (20 minutes)

Teams perform their scripted stories enhanced by robot interactions, showcasing both their dramatic work and technological skills.

Following the performance, teams share reflections on their learning journey, discussing challenges, successes, and insights gained throughout the project.

Differentiation notes:

- Flexible performance roles are created to accommodate varying comfort levels among students.
- Alternative participation options are provided for students experiencing performance anxiety.
- Recognition emphasizes and celebrates diverse contributions and individual strengths within each team.

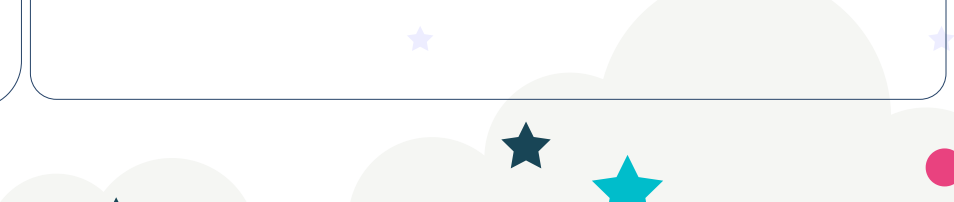
Closing (10'):

Class Reflection and Discussion

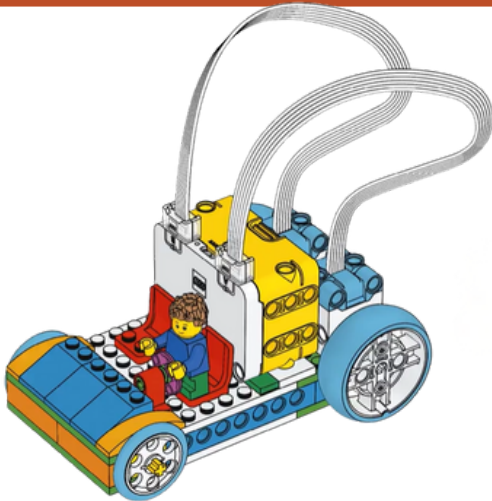
For Students:

- Review the project journey timeline to recap key milestones (“Were there any unexpected changes or delays? How did we handle them?”).
- Participate in peer appreciation activities to recognize classmates’ contributions (“How did your peers’ efforts impact the success of the project?”).
- Share learning highlights and memorable moments from the project (“Was there a moment during the project that you found especially rewarding or fun?”).

For Teachers:

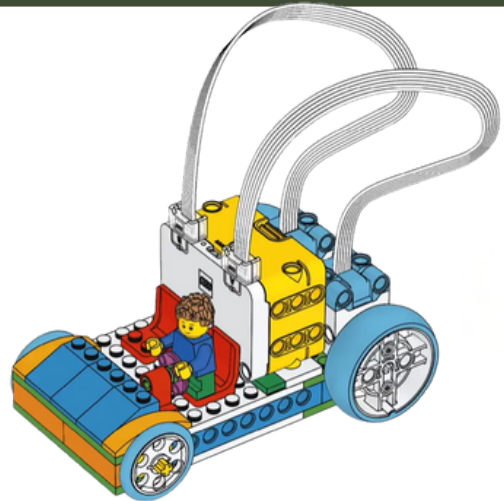
- Record implementation notes reflecting on what worked well and potential improvements.
 - Monitor and document student engagement throughout the project.
 - Collect evidence of cross-curricular connections made during the activities for future reference.
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Annex: Robot movement cards



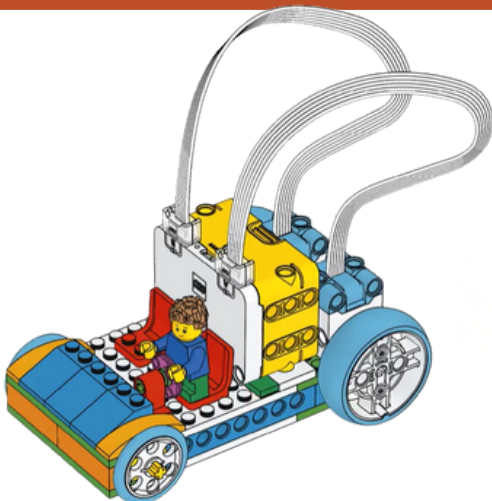
■ **Challenge 1: Move It!**
Make your robot move forward 30 cm and stop.

☞ **Hint:** Use the "Move motors" block and test distance.



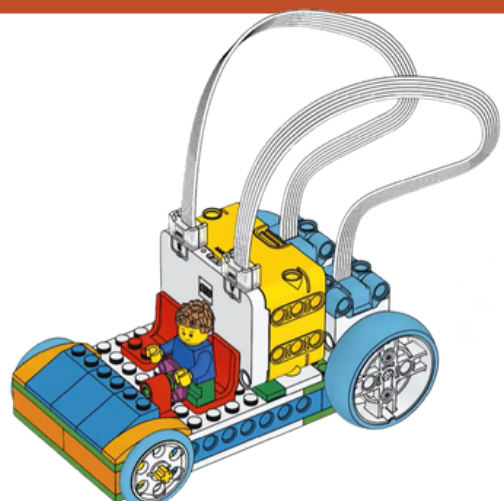
■ **Challenge 2: Spin It Around**
Make the robot turn in a full circle (360°).

☞ **Hint:** Experiment with turning one motor forward and the other backward.



■ **Challenge 3: Maze Runner**
Navigate a simple maze with at least 2 turns.

☞ **Hint:** Break movements into steps and use precise turns.



■ **Challenge 4: Tilt to Go**
Tilt the robot forward to make it move, and tilt it back to make it stop.

☞ **Hint:** Use the pitch angle from the movement sensor to control motion.

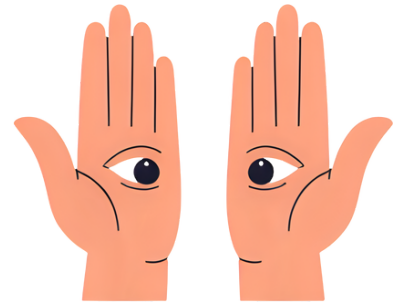
Annex: Phrase cards



**The moon whispered
secrets to the sea.**



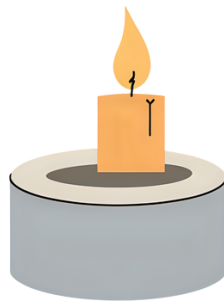
**Where there is
smoke, there is fire.**



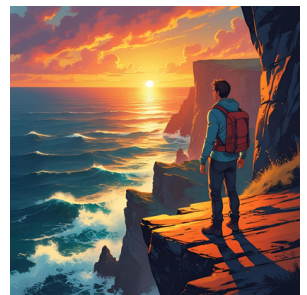
**What the eyes hide, the
hands reveal.**



**A heart wrapped in
silence beats the
loudest.**



**Even the smallest
candle pushes back
the night.**



**Fortune favors the
bold.**

Annex: programming vocabulary reference sheet

Programming Vocabulary Reference Sheet - LEGO SPIKE

(For Elementary Students)

 Command Name	 Block Icon / Description	 What it Does
When program starts	Essential block to start	Starts the program
<i>Move Forward</i>	<i>Move motor for X cm forward</i>	<i>Moves the robot forward</i>
<i>Move Backward</i>	<i>Move motor for X cm backward</i>	<i>Moves the robot backward</i>
<i>Turn Left/Right</i>	<i>Turn 90° or One motor only</i>	<i>Turns the robot</i>

Student Tip:

- ✓ Read what each block does before using it.
- ✓ Keep this sheet nearby as a quick reference.